Question	Answer	Marks	Partial Marks
4(a)	246	**3**	**B2** for *BCS*(outh) = 66 or *BCA* = 48 **and** *ACN*(orth) = 66 or *BCW*(est) = 24 or *ACS*(outh) = 114 or **B1** for *ABC* = 66 or *BAC* = 66 or *BCA* = 48 or *ACN*(orth) = 66
4(b)(i)	58	**1**	
4(b)(ii)	106	**1**	
4(b)(iii)	47	**2**	**B1** for *PRQ* = 27 or **B1FT** for *SPR*, either = 48 or = 106 – *their* **(b)(i)** or **B1FT** for *RPQ* = *their* **(b)(i)** – 11
4(c)	Radius perpendicular to tangent	**1**	
	Tangents to circle from a/same point oe	**1**	
	RHS	**1**	
	68 angles on a [straight] line add up/sum to 180 oe	**1**	
	56 [base angles of] isosceles triangle	**1**	
	OBC = *BOT* Alternate angles	**1**	Angles and reason required and dependent on *OBC* and *BOT* correct
5(a)(i)	2[.00] or 2.002 to 2.003 nfww	**3**	**M2** for $\sqrt{4.8^2 + 5.6^2 - 2 \times 4.8 \times 5.6 \times \cos 20.4}$ OR **M1** for $4.8^2 + 5.6^2 - 2 \times 4.8 \times 5.6 \times \cos 20.4$ **A1** for 4.01[17…] or 4.012
5(a)(ii)	4.1[0] or 4.11 or 4.100 to 4.107 cao	**2**	**M1** for $\tan 64 = \dfrac{AX}{\textit{their}\ \textbf{(a)(i)}}$ or for $\dfrac{AX}{\sin 64} = \dfrac{\textit{their}\ \textbf{(a)(i)}}{\sin(90-64)}$ oe
5(a)(iii)	6.96	**2**	**M1** for $\frac{1}{2} \times 4.8 \times 2.9$ oe

Question	Answer	Marks	Partial Marks
5(b)	11.3 or 11.31..	**5**	**M4** for $2\times\frac{8}{\sin(45)}\times\sin 30$ or **B4** for *PM* = 5.65[685...] or 5.66 or better OR **B1** for *angle RPM* = 45° **M2** for $\frac{8}{\sin(their\ 45)}\times\sin 30$ or **M1** for implicit form
6(a)(i)	Correct curve	**3**	**B1** for correct horizontal placement for 6 plots **B1** for correct vertical placement for 6 plots **B1 dep on at least B1** for reasonable increasing curve through *their* 6 points If 0 scored, **SC1** for 4 out of 6 points correctly plotted
6(a)(ii)(a)	87 to 89.5	**1**	
6(a)(ii)(b)	12.5 to 14	**2**	**B1** for [LQ =] 80.5 to 81.5 or [UQ =] 94 to 94.5
6(a)(ii)(c)	Strict FT, 200 – *their* cumul freq reading from *their* graph at 110 given to nearest integer	**2**	**B1FT** for correct cumul freq at 110 seen or for non-integer answer
6(b)(i)	3576	**4**	**M1** for midpoints soi **M1** for use of $\sum fx$ where x is in the correct interval including boundaries **M1** (dep on 2nd **M1**) for $\sum fx \div 50$
6(b)(ii)	5 3.2 3	**3**	**B1** for each If 0 scored, **SC1** for 3 frequency densities $\frac{12}{600}, \frac{15}{900}, \frac{16}{1500}, \frac{7}{700}$ seen oe to 3sf or better or multiplier 3 or 300
7(a)	Cubic	**1**	